DESIGNING AN AUTOMATED LIBRARY USER REGISTRATION SYSTEM FOR NKUMBA UNIVERSITY LIBRARY

BY

PHIONAH AGIREMBABAZI

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215020767

SUPERVISOR

DR. JOYCE BUKIRWA

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ABSTRACT

Due to the fact that Nkumba university library is using the manual system to register the library users and mainly it is a paper based system, such records are expensive to store. They are also easy to destroy and difficult to retrieve. Therefore the study aimed at designing an automated library user registration system for Nkumba university library which will help to improve efficiency, productivity and generally good services to the library.

This research project was carried out at Nkumba University under the library department and it aimed at designing an automated library user registration system for Nkumba university library.

The research report consists of six chapters;

Chapter one; it consist of background of the study, statement of the problem, scope, aim and objectives of the study, research questions, justification of the study and definition of key terms.

Chapter two: it's the literature review and it is the related literature that is published according to the topic

Chapter three: it's about the research methodology, it contains the area of study, research design, population of study, sampling, data collection methods which include the questionnaire, interview of which the research instruments are; questionnaires and interview guide. It also has ethical issues and limitations o the study.

Chapter four: it includes the analysis and presentation which include findings and data was presented in numbers using tables.

Chapter five: includes designing a new system, a user requirement that is functional of the proposed system and non- function for the automated system.

Chapter six: includes summary of findings, conclusion and the recommendations of the project

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DECLARARION

I Agirembabazi phionah declare that this work is as a result of my research and nothing in this form has ever been submitted any institution for award.

Signature:…… Date:…….

AGIREMBABAZI PHIONAH

APPROVAL

My work has been developed under the guidance and supervision and it is worth to be submitted to Makerere University for the award of bachelor's degree in library and information science.

Supervisors Name	Signature	Date
DR.JOYCE BUKIRWA	……..	……

DEDICATION

This project report is dedicated to my parents, husband, brothers and sisters and my course mates for the support that cannot be exhaustively defined. May God's blessings be multiplied to you all.

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I thank the Almighty God who cleared my way and gave me wisdom to study.

Special thanks to my supervisor and head of department Dr. Joyce Bukirwa for guiding me throughout the study period. This task would not have been a success without her technical assistance. Thank you so much Doctor

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I also extend my appreciation to the employees of Nkumba university library. Thank you all for the support and all the cooperation you rendered to me that helped me to accomplish my project.

GOD BLESS YOU ALL

LIST OF ABBREVIATIONS

HTML : Hyper Text Markup Language

MSQL **:** My Structured Query Language

PHP **:** Hypertext Processor (Personal Home Page)

EASLIS : East African School of Library and Information Science

CPU **:** Central Processing Unit

ORNSIS: Online Registration New Student Intake System

ID NO. :Identification Number

RAD: Rapid Application Development

SDLC: Systems Development Life Cycle

RFID: Radio frequency identification

CHAPTER ONE: INTRODUCTION TO THE STUDY

1.0.Introduction

This chapter introduces us to the background of the study, statement of the problem, purpose of the study, objectives of the study, justification of the study and the key definition terms.

1.1. Background of the study

1.1.1. Conceptualization

An academic library is the heart of a university and this is because it serves as hub for all academic activities in the university. It serves various functions including: supporting the parent organization in achieving organizational objectives ,provide support for research and educational activities in universities, institutions and other academic institutions, and also serve the unique teaching, learning and research needs of its home academic institution by being active participants in the creation, transmission and dissemination of knowledge. Academic libraries serve users like; students, lecturers (staff) and external users. The library users need to register with the library in order to become the potential users. (Fleming, 2010).

Academic library is a component part of the history of civilization and has a responsibility to its environment hence the basic philosophy of the university library is based on the concept of service and provision of relevant information materials for users. The information materials include books, journals, documents, e-resources and audiovisual materials (Butller&Lorrie, 2007).

Libraries are referred to as growing organism and the growth can only be seen in terms of materials, staff and users. The population of users therefore, must be taken into consideration so that the library will be guided while making decision about collection development for scholarship provision of other learning resources and facilities such as furniture and other equipment. Users' registration is thus, the only way to determine the users' population in real time. User registration as the process of establishing a data bank of information about library users .This information is useful when a user wants to make use of library materials outside the library or requesting for inter-library services (Lorrie, 2007)

1.1.2. Background to Nkumba University

Nkumba University is a private university that started 24years ego. It evolved from a kindergarten, college then to a university in 1994. It was granted a charter in 2006. It comprises of seven schools which include,

1) School of business administration(SBA)
2) School of education humanities and sciences(SEHS)
3) School of commercial, industrial art and design(SCIAD)
4) School of hospitality and environmental science(SHES)
5) school of law(slaw)
6) school of social sciences(SOSS)
7) School of postgraduate studies and research(SPSR)

Nkumba University has a four storied library building which has been in existence since 1995. With each level specialized for certain information material for example;

a) ground floor for law books
b) first floor for education, business social sciences books, journals and magazines
c) second floor for postgraduate and question papers for all courses
d) third floor for nursing books and dissertations

Nkumba university library is charged the task of supporting all academic programs of the university through provision of adequate, accurate and relevant information to support the learning teaching and research process of the university and it exists to serve the clientele of 3000 users including students (2700), staff (250) and 50 of the wider community **(Nkumba university library strategic plan, 2008/2012)**.

1.1.3. Vision of the library:

To provide access to the words knowledge in order to promote and support the teaching, learning and research activities of the university and the community.

1.1.4. Mission:

To plan, avail and facilitate access to a broad spectrum of information resources and services in an outstanding learning environment with friendly and expert help.

1.2. Problem Statement

Nkumba university library is using the manual system for registering library users .The problem associated with using such a system is that there is a risk of data loss resulting from actual physical damage of the records for example by fire, water ,paper pluck outs to mention but a few. Also too much time is spent on capturing the library users information hence have to wait for so long, storage space and maintenance of such records including preservation is labor intensive and costly .this kind of system causes storage and retrieval of users' records difficult when it comes to producing a list of defaulters which therefore calls for designing an automated library user registration system. It has also been observed that there are seven schools in Nkumba University leading to increasing population of students of which one of the requirements of being Nkumba university student is to register with the library. Basing on this problem statement therefore, there is need to design an automation system for registering library users for Nkumba University library.

1.3. Aim of the study

The aim of the study is to design an automated library user registration system for Nkumba university library.

1.4. Specific Objectives

1) To identify the library users of Nkumba university library.
2) To examine the current system used in registering users in Nkumba university library.
3) To identify the gaps/challenges with the current system of user registration in Nkumba university library.
4) To identify the requirements for a user registration.
5) To design an automated system for registering library users in Nkumba university library.

1.5. Research Questions

1) Who are the library users of Nkumba university library?
2) What is the nature of the current system used in registering users in Nkumba university library?
3) Which challenges are faced with the current system of user registration in Nkumba university library?
4) What requirements are necessary for an automated user registration system?
5) What is the suitable design for an automated system for registering library users in Nkumba university library?

1.6. Scope

1.6.1. Geographical Scope

Nkumba University is located in Nkumba Hill in Wakiso District, approximately 10 kilometers (6.2 mi), by road, north-east of Entebbe, International Airport along the northern shores of Lake Victoria, the second-largest fresh water body in the world.

1.6.2. Content

This study intended to design an automated library user registration system for Nkumba university library. Data collections covered the library staff, library users, the system used in the library, challenges faced by the library staff in registering library users.

1.7. Significance of the Study

I. The project leads to an improved and organized library user registration system over the manual system of Nkumba university library.
II. It helps to reduce on the work load of registering library users in books since the automated system will be assessed hence minimizing the risks of record misplacement.
III. The study helps to identify user requirements like accessing the internet and other library services.

IV. It also help to enable users accept the change from traditional library system to modern/digital management system especially the automated library user registration system.

V. The proposed system intends to have a centralized database for all library users which will reduce data redundancy and data inconsistency especially about the library users. This enables Nkumba university librarians to effectively manage library users' records /data in the library.

VI. The system also reduces operational delays in providing relevant services since it will be automatic and very fast hence saving both librarians and users time.

VII. This is a source of literature to be viewed by the forthcoming researchers that might be interested in revealing more knowledge about this topic hence contribution to knowledge.

1.8. Definition of key terms;

1) **Automation:** this is the use of computer-assisted systems to undertake repetitive tasks leaving staff free to carry out other, more satisfying duties(Breeding,2009)
2) **Academic libraries:** these are integral part of colleges / universities and have a critical role to play in supporting the core mission of the university which is teaching, learning and research(editors;Musoke&Kakai,2006)
3) **Registration:** the process examining readers application forms and making out membership ticket or card OR the act of entering information about something in a book or computer(lynch,2014)
4) **Library automation:** it is the use of computers and networking technologies in the library (Lam,2002)
5) **User registration,** this is the process of recording library user information for example names registration number student number and others so that he or she becomes eligible to use the library(Buwule,2017)
6) **Library** as a collection of sources of information and similar resources, made accessible to a defined community for reference or borrowing(Buwule,2017)

7) **Library:** is a collection of books and other literary material kept for reading, study and consultation OR it is a place ,building, room set apart for keeping and use of a collection of books (Montague,1995)

8) Library users. These are registered users who can be able to use the cards given to them to access information materials in the library(Buwule,2017)

CHAPTER TWO: LITRETURE REVIEW

2.1. Introduction

This chapter summarizes the relevant literature according to different scholars basing on the research topic.

2.2. Library Users in Academic Libraries

Academic libraries are established for teaching and providing services to the institutions and professionals. Librarians in academic libraries are teachers and they teach and provide access to information to users and are often consulted by students when preparing for term papers and doing literature searches (editor; Wamunza, 2004).

In this new technology environment, professional librarians are required to mould what is referred to as academic user of the 21st century and the user needs to posses the skills, knowledge and attitude to fully exploit the wide range of electronic resources available. (Wamunza, 2004)

Choo Ming(2000:26) argues that the academic users of the 21st century need both computer and information literature and the richest online information is not beneficial unless they know what resources are available and how to extract required information from appropriate sources.

According to Ying (2002:3), a library reader or user of the 21st century is identified into five qualities;

1. Active aspirants after knowledge information
2. Active users of the library
3. Experts in applying modern technology
4. Conscious participants in library management
5. Information literates. Basing on these five qualities therefore, librarians should be aware that it is their professional duty to mould the academic library users of the 21st century by developing innovative information literacy programs to train library users on new skills.

In academic libraries, in order to understand how the concepts of the library, computer and information literacy impact the learning of university students “as users in academic libraries”, one needs to know the educational characteristics of students and to situate and relate it to their

educational backgrounds especially in terms of library resources implying that they have underdeveloped reading culture (Wamunza, 2004:56).

Wamunza (2004:76) notes that teaching staff as library users in academic libraries need to acquire new skills to work with electronic information environment and to integrate that into their teaching. The Association of college and research libraries(ACRL) has also stressed that the teaching staff should establish the Context for learning ,inspire students, after guidance for fulfilling information needs and monitor students progress and they should be ready and prepared to integrate information literacy aspects in their teaching so that students are taught viable information skills of accessing information resources.

A lot of information can be retrieved using electronic resources both inside (students and staff) and outside (registered external readers or users). This may be the reason why they are more popular compared to the resources. However, lack of computer and IT skills, time consuming, limited access to computer terminal and too much information retrieval using electronic resources, often distracts from doing work. These challenges among users affect them from using electronic information sources (Jayaprakash, 2011).

Nkumba university: library and information services handbook(2012/2015) states that Nkumba university library has two categories of users who include; internal users (staff and students) and the external users who are allowed to use the library materials including the internet upon paying user fees for example,

i. one day a local external user pays three thousand(shs3000) and an external international user pays twenty thousand shillings(shs20000)
ii. One week local pays ten thousand(shs10000)
iii. Two months local pays forty thousand (shs40000) to mention but a few.

2.3. Library User Registration: The Need for Registration

The process of library user registration involves capturing or retrieving information about a specific student, staff or external reader. It is necessary to register library users in order to;

1. Increase accessibility of information to the registered library users
2. To maintain records in right custody
3. To ease prompt preparation of accountability reports
4. To serve as managerial tool which strengthens the academic libraries' performance hence need for registration (Saleem, 2009).

According to Nkumba University: library and information services handbook (2015), registration is very important for many users as stated below;

1) Saves time in the library once the user is registered; it takes few minutes to access any information material required.
2) Eases accessibility since the user has a library card, it is easy to enter the library without any interruption from the security desk
3) Promotes user satisfaction because librarians able to lend a hand as fast as possible.

Academic libraries serve their patrons, staff and other registered users (external readers) by fulfilling seven basic roles including;

1. Knowledge archival
2. The preservation and maintenance of culture
3. Knowledge dissemination
4. Knowledge sharing
5. Information retrieval
6. Education
7. Social interaction. Each of these roles offers the above library users the opportunity to recognize and view libraries as integral part of democratic society where access to free information has been both expected and demanded (Lorrie, 2007).

2.4. Challenges Associated With User Registration

Registration of library users in academic libraries put too much pressure on library staff to be correct in all details of their work at all times, the problem is no man is perfect and this is mainly associated with the use of manual registration system(Bananuka,2005).

According to Bananuka (2005), the challenges associated with library user registration include the following;

1. Security threats such as data manipulation by the authorized users and loss of data are always common with the manual system of registration.
2. Time consuming and slow due to the use of paper while registering users manually and the fact that the library staff has to go through several paper works and verification of each individual which demands for ample time.
3. Establishing defaulters which involves calculating overdue fines and enforcing policy using policy using manual is prone to mistakes.
4. Too much paperwork, paper used all the time when doing work generates a lot of records and keeping them can be a problem in a sense that they can be mixed up when making retrieval difficult especially when reports are urgently needed for decision making. There is also a possibility of losing papers or misplacing them yet contains vital information which is used to monitor book bank activities.
5. Duplication and redundancy of data which is caused by increasing size of data due to growth of documents and resources which makes the system and the many codes variants hard to understand and cause maintenance headache.

2.5. Requirements for the Library User Registration

According to the Nkumba university: library and information services handbook (2015), to register an academic library reader/user, one has to turn to the library information desk with the necessary documents (admission letter, bank slip, valid university identity card).this can be done using two systems;

2.5.1 Manual System

The library staff gets the book and takes recordings as necessary and the patron is thereafter given a library card which he or she fills name, student number and then stamped.

The most common manual system in use is the Newark system and its variation where libraries use a ledger system to record their users. After the users details are captured in the ledger, a valid library card is provided which is meant used for identification and borrowing purposes. It involves the insertion of a circulation card with call number, author, title in a paper pocket in each item (Laitatul, 2007).

2.5.2 Automated Library System

An electronic library card registration allows patrons to self register for the library card online, remotely or onsite, adding convenience for the patron and time saving for staff. One of the kinds of online library registration is E-CARD which verifies and standardizes the address information entered by the patron in real time. It includes the following features;

1. Online and in branch registration
2. Address verification
3. Name verification. (Butter &Horning,2007)

2.6 Study Gap

The above literature has been reviewed by the different scholars about the automation of the library user registration but in context of Nkumba university library, no study has been conducted in order to come up with the automated library user registration system that is why the researcher is interested in this study.

CHAPTER THREE: RESEARCH METHODOLOGY

3.1. Introduction

This chapter presents the area of study, research design, population and sampling, data collection methods and instruments, data presentation and analysis and system testing.

3.2. Area of Study

The study was conducted in Nkumba university library with the major aim of designing an automated library user registration system. Nkumba University library is located in Nkumba University on Nkumba hill wakiso district north east of Entebbe international airport along the shores of Lake Victoria.

3.3. Research Design

Research design is the framework for the collection and analysis of data. It is also known as the conceptual structure within which research is conducted (Bryman, 2008).

Case study was the research design used in this study. Case study is instrumental and focuses on a particular phenomenon or theory (Pickard, 2007).

In this case the researcher looked at library user registration for Nkumba university library

Surveys of respondents were carried out using questioning and interviewing and the respondents were students and staff.

In data collection, various methods were used to collect data which included; interview, questionnaires, document analysis

3.4. Study Population

Population refers to group f individuals, persons, objects or items from which samples are taken for measurements (Bryman, 2008)

Students and staff were the target population since they are the daily users of the library. It helped to collect accurate data.

3.5 Sampling

Sample size is a finite part of a statistical population whose properties are studied to gain information about the whole selected from a larger population for the purpose of survey (Bryman, 2008).

Sample size was provided basing on the respondents in questionnaires and interview guide.

The sampling technique used is purposive which helped to conduct the study and acquire knowledge about the user needs in the system. With this strategy, the researcher focused on the library staff of Nkumba University about the current system they use. In this strategy, the researcher also collected data from students.

3.6. Data Collection Methods and Instruments

Data collection refers to the process of using some methods (Interview, questionnaire, observation) to collect data (Powell, 2004)

Instrument refers to a tool that can be used to collect data for example; questionnaire, interview guide (Powell, 2004)

The following are the data collection methods and their respective instruments which will be used;

3.6.1. Interview

An interview is a conversation or dialogue with a purpose between two people. It involved asking questions and getting answers from participants in the study. Interview helped to give accurate information on the current system used and enable the researcher get final results.

The interview guide is the instrument that the researcher used to obtain information from the library staff of Nkumba University. The researcher interviewed the library staff since they are the ones with relevant information mostly about who the library users are, current system used to register users and challenges encountered by both library staff and library users when using the system.

3.6.2. Questionnaire

Questionnaire refers to the type of data collected where the researcher sets questions to be answered by the respondent. It is both structured and unstructured. It will be used to set questions to collect information from the library staff and students.

It is referred to as a data collection tool in which there are written questions to be answered and presented to the respondents in written or typed form (Chaleunvong, 2009)

The instrument used was the questionnaire.

3.6.3. Document Reviewing

This refer to the analysis of documents that contain information about the public, with this method of data collection, the researcher reviewed a number of statistical books and articles about automation ,academic libraries, system .Such documents provided primary and secondary data that helped the researcher get information on how to design the automated library user registration system.

3.7 Data Quality Control

Data quality refer to a set of procedures intended to ensure that information produced adheres a defined set of quality criteria or meets the requirements of customer o client(Kothari,2012)

Data quality control was done to check the validity and reliability of the instrument by carrying out preliminary tests on the interview guide and questionnaire.

3.8 Data Analysis and Presentation

Data analysis refers to a process of working with data to describe, discuss, interpret, evaluate and explain data in terms of the research questions or hypothesis of the research project (Matthew, 2010).the data was quantitatively analyzed, all data collected was edited to check for its completeness and accuracy data was analyzed to establish the percentage and frequencies.

Data presentation is defined as presenting the findings which available to the interested parties (Mugenda, 2003)

Data was collected, analyzed and tabulated. This helped to check the accuracy, completeness, comprehensibility and legibility of the answers that were given and editing was done before tabulation.

3.9 Ethical Considerations/Issues

This helped to conduct the study in a systematic way, honesty and integrity throughout the process and the respect of confidentiality of the respondents

3.10 Limitations of the Study

The researcher encountered the following limitations;

Financial problem. The study needed a lot of money for many functions for example printing travelling to the field, feeding hence a limitation.

Attitude of the respondents was another problem in that some respondents were reluctant to give adequate information hence time wasting

Time management. Time allocated to the researcher was not enough to gather information from the field and interpret it. However the researcher tried the level best to do work basing on the work plan.

CHAPTER FOUR: PRESENTATION, ANALYSIS AND DISCUSIION OF FINDINGS

4.0 Introduction

This chapter presents the data gathered from the field and analyses it basing on the research questions as follows;

1) Who are the library users of Nkumba university library?
2) What is the nature of the current system used in registering users in Nkumba university library?
3) Which challenges are encountered in using the current system of user registration in Nkumba university library?
4) What requirements are necessary for an automated user registration system?
5) What is the suitable design for an automated system for registering library users in Nkumba university library?

4.1 Research question 1: who are the library users of Nkumba university library?

From the research question one, the researcher using interview guide and questionnaire method asked the library staff and library users "who are the library users of Nkumba university library?"

Findings from the library staff indicated that an academic library provides access to knowledge, information through a range of resources and services equally available to all registered library users regardless of sex, nationality, religion, language, disability, educational attainment.

Findings from library staff revealed that currently the number of people who use the library has immersevely grown and therefore there is need and request from users to install internet services to solve the problem of overcrowding. However, this is a challenge because it requires enough funds to install internet services

Library users through the questionnaires answered that other library users are; researchers, lecturers, registered external users.

4.1 Response Rate

Table 4.1 Response rate

Respondents	Actual	Percentage (%)
27	**73**	**36.9**

Source: primary data, 2018

From the above table 4.1, the researcher found out that of the 73 Key respondents she had initially targeted 27 provided successful responses which indicate that 36.9% of the respondents provided relevant data which renders the finding credible and reliable for the study.

4.2 Respondents Personal Variables

Table 4.2 Respondents' sex of respondents

Gender	Frequency	Percentage (%)
Male	**15**	**55.6**
Female	**12**	**44.4**
Total	**27**	**100**

Source: primary data, 2018

From the table 4.2 above, it was discovered that of the 27 respondents who provided data for the study 15 were males 12 females. This was so because more males were willing to take part in the study. But both male and female provided relevant information for the study

4.2 Research question 2: The Nature of the current system used to register library users in Nkumba university library

Regarding the type of system used in Nkumba university library to register users, respondents were asked to identify whether it was "manual or automated system"

Table 4.2.1 description of the current system used to register library users

The nature of the current system used	Frequency	Percentage(%)
Manual	**27**	**100**
Automated	**00**	**00**
Total	**27**	**100**

Source: primary data, 2018

Findings regarding from library staff, students mostly agreed that the manual system is used to register its users that is flexible

One staff revealed to the researcher that since its establishment, registration has been done manually on cards and register books although one revealed that the automated system is in place but not functional due to low internet connectivity and lack of funds to monitor it.

The respondents further revealed that the manual system does not adequately serve the needs of library users because it cannot address the issue of increasing number of users and limitations in information dissemination and security materials.

The researcher therefore argues that manual system cannot effectively serve the needs of users and therefore automating the registration system will to a great extent quicken the process of information provision in Nkumba university library because the extent to which the library satisfies users information needs is fundamentally more important .this is because the ultimate goal of academic libraries to bring about higher users satisfaction

4.3. Research question 3: Which challenges are encountered in using the current system of

User registration in Nkumba university library?

Respondents were asked; which challenges are encountered in using the current system? Among the challenges noted include the following as stated in the table below;

Table 4.3.1 challenges encountered in using the current system

Challenges encountered in using the current system	**Frequency**	**Percentage (%)**
Pressure	**01**	**3.7**
Time consuming	**12**	**44.4**
Data redundancy	**02**	**7.4**
Too much paperwork	**12**	**44.4**
Total	**27**	**100**

Source: primary data, 2018

One of the library staff said the manual system they are using puts pressure on them to be correct in all details of their work at all times an with the nature of the system, the level of services depend on individuals which puts a requirement on management to put training continuously on staff to keep them motivated and to ensure they are following the correct procedures.

12 library staff also said that the manual system is time consuming while registering users which delays service delivery

Another challenge mentioned was too much paper work used hence if not in good storage may leave information misplaced.

Another respondent mention of data redundancy due to the manual system used to register users.

4.4. Research problem 4: What requirements are necessary for an automated user registration system?

4.4.1.Functional Requirements

Functional requirements describe a systems services or functions. This is how a system should react to certain inputs and how it should behave in certain circumstances. For a user registration system, it should be able to identify every user uniquely the system should make adequate provisions for search facilities on the database.

4.4.2. Software And Hardware Requirements

There are various hardware and software that are required to effectively develop a user registration system. Hardware requirements include; a computer set as CPU, monitor, keyboard mouse memory space. Software requirements include, web server visual studio for designing webpage interface, Adobe Photoshop CS7.0 for creating buttons and altering images in the web page, network requirement such as internet line to mention but a few.

4.5. Research question 5: What is the suitable design for an automated system for registering library users in Nkumba university library?

Chapter Summary

Chapter four represents the research findings on designing an automated library user registration system for Nkumba university library. The findings have been presented using narrative, frequencies, percentage and tables. The findings reflect the true findings according to responses on questions provided in the questionnaire and interview guide.

CHAPTER FIVE

5.0. System Analysis and Design

This comprises the study of the existing system, weaknesses of the current system specifications of the functional and nonfunctional system requirements, user requirements of the new system and among others.

5.1. The Proposed Automated System

The automated library user registration system will enable to perform the following tasks electronically;

- Addition, removal, edition of library user profiles
- Keep tracking of daily library user registration
- Keep tacking of daily arrival/departure time of library user in the library
- It will help to generate a number of reports daily ,weekly and yearly basis

5.2. Technology

This section describes a technology that has been researched and used for development of this project. Coding is the most common ID technology. It is an automatic identification (Auto ID) technology that streamlines identification and data collection. The benefits of using Barcodes for automated data collection are very simple: speed and accuracy. Time after time, it has been proven that entering Barcode data is at least 100 times faster and more accurate than traditional manual keyboard entry, which translates into a dramatic increase in efficiency and productivity for any operation [http://www.bsigroup.com/en/ProductServices/Medical/QMS-for-medical-devices-ISO13485/What-do-I-need/)].

A barcode is series of bars and spaces arranged according to a set of rules that determines how data is to be represented. Different bars and spaces patterns are used to express different symbols. These symbols are readable only by a scanner. Barcode technology is an important identification tool that provides an accurate and timely support of the data requirement for proper management systems. The mapping between messages and barcodes is called symbology. The

specification of a symbology includes the encoding of the single digits/characters of the message as well as the start and stop markers into bars and space, the size of the quit zone required to be before and after the barcode. There are over a hundred different barcode symbologies invented. However, only a half-dozen are used regularly (They are Code 39, Codabar, U.P.C., Interleaved 2 of 5 and Code 128). Barcode symbology types include: (1) linear barcodes such as EAN-13, Interleaved 2 of 5 (Code 25, I2of5, ITF, I25), and Code 39, and 2-D barcodes such as MaxiCode, PDF-417, and DataMatrix. Figure.1 illustrates examples of different types of barcodes including linear symbologies, and 2-D symbologies (refer Figure 1).

In libraries in Bangladesh [Robertson & Jalaly, 2003; Angeles, 2007], they use the barcode technologies to automate the data-capturing process, for stock verification, and in the circulation counter for issue/return of documents the statistic of staff that using the technology is shown in the Figure 2. The barcode technology can play a significant role in the automation of library management systems in spite of having some limitations of the technology [Kern, 2004]. This technology is an appropriate technology for the management of learning resources and library services. Ensuring uncorrupted and error free network systems and uninterrupted supply of electricity can surpass the limitations of barcode technology. Although radio frequency identification (RFID) systems have been in use in libraries for five years for book identification, self-checkout, anti-theft control, inventory control, and sorting and conveying library books and AV materials [Rajendiran, & Bhushan, 2006], Bangladesh has not yet started to use RFID in libraries.



Figure 1: Different types of barcodes: (a) EAN-13, (b) interleaved 2 of 5 (Code 25, I2of5, ITF, I25), (c) Code 39, (d) MaxiCode and (e) PDF-417.

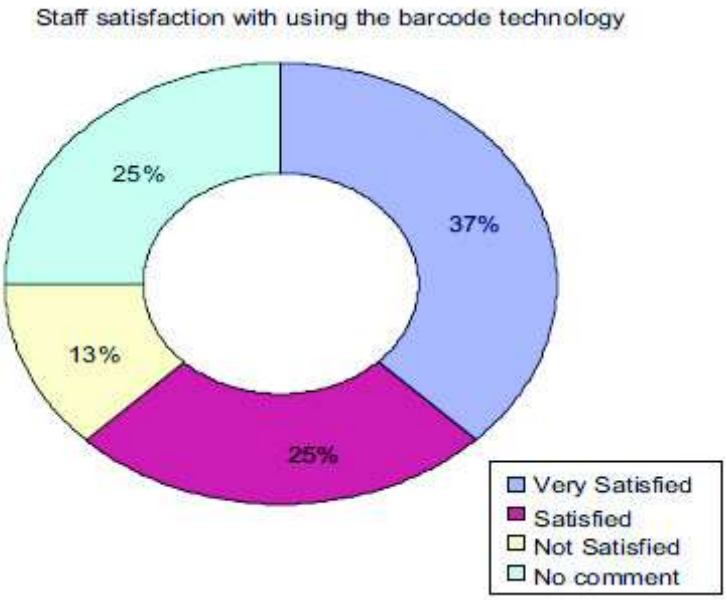


Figure 2: Staff satisfaction with using the barcode technology in the surveyed libraries [http://sgsportal.upm.edu.my/portal/page?_pageid=116,1,116_34141&_dad=portal&_schema=PORTAL,].

5.3. Methodology

Methodology is a guideline for a developer to structure, plan and control the process of development a system.

To develop the application or software various methods can be used. Choose the right and suitable methodology is to ensure the project success. The Systems Development Life Cycle (SDLC) is a conceptual model used in project management that describes the stages involved in an information system development project from an initial feasibility study through maintenance of the completed application. Various SDLC methodologies have been developed to guide the processes involved including Rapid Application Development (RAD). Figure 3 shows the RAD phases.

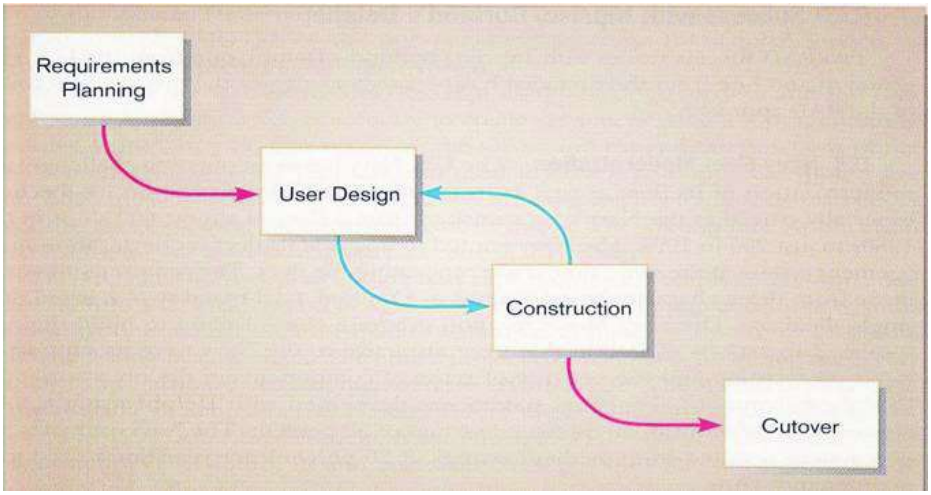


Figure 3: RAD phases

5.4. Online Registration New Student Intake System (ORNSIS) in the library

This section presents the ORNSIS design and interfaces. In ORNSIS the user interface have been design; in Figure 4 the welcome page, user need to insert username and password then click login. The welcome page will bring user to accept offer page if user can go through welcome page with correct username and password.

Below is a screenshot of a welcome page;

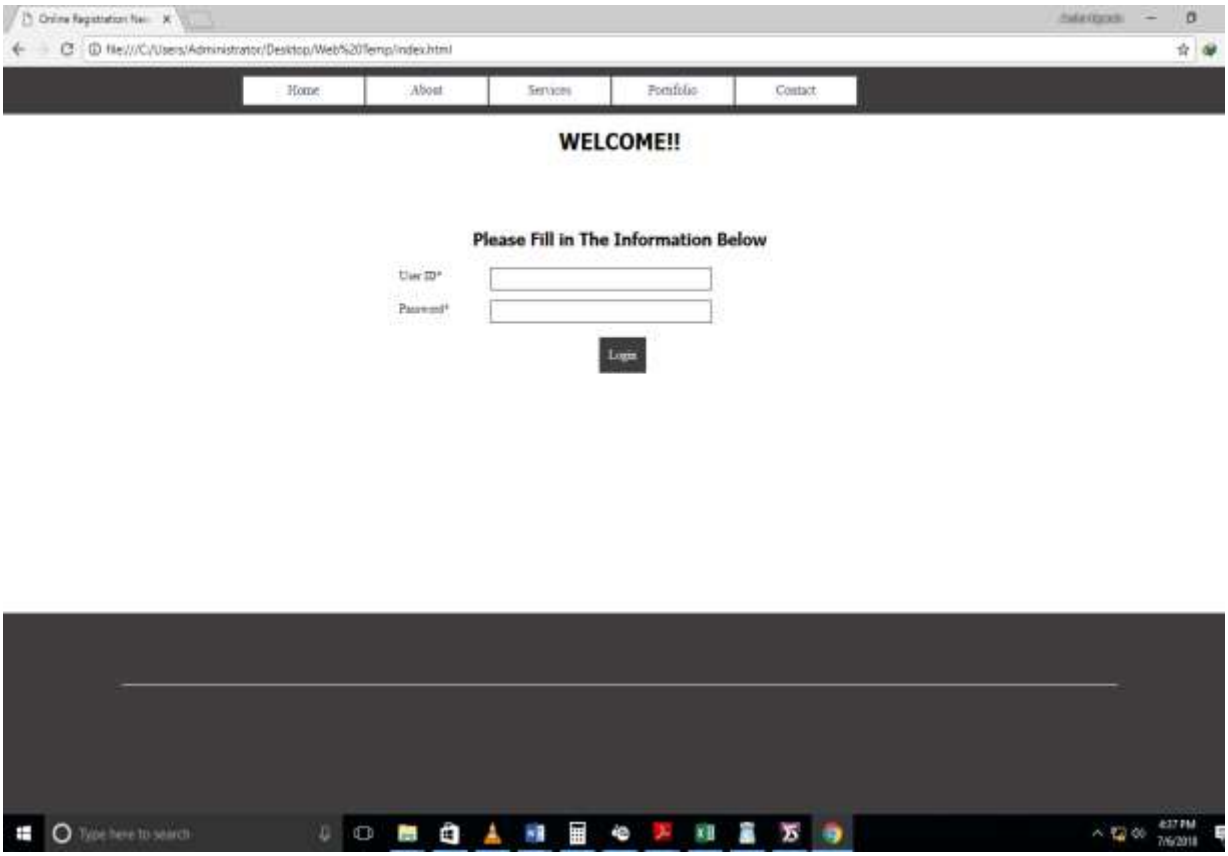


Figure 4: Welcome page (login page).

User will go through the personal information form. In personal info, user can view and change necessary information and submit it as to save the info (see Figure 5). Then, the user will go through to Accept offer form need user to read the term and condition carefully and click to select the checkbox to ensure user understand with term and condition (see Figure 5). Then user need to choose whether to choose YES or NO for accept offer.

Screen shoot for personal information.

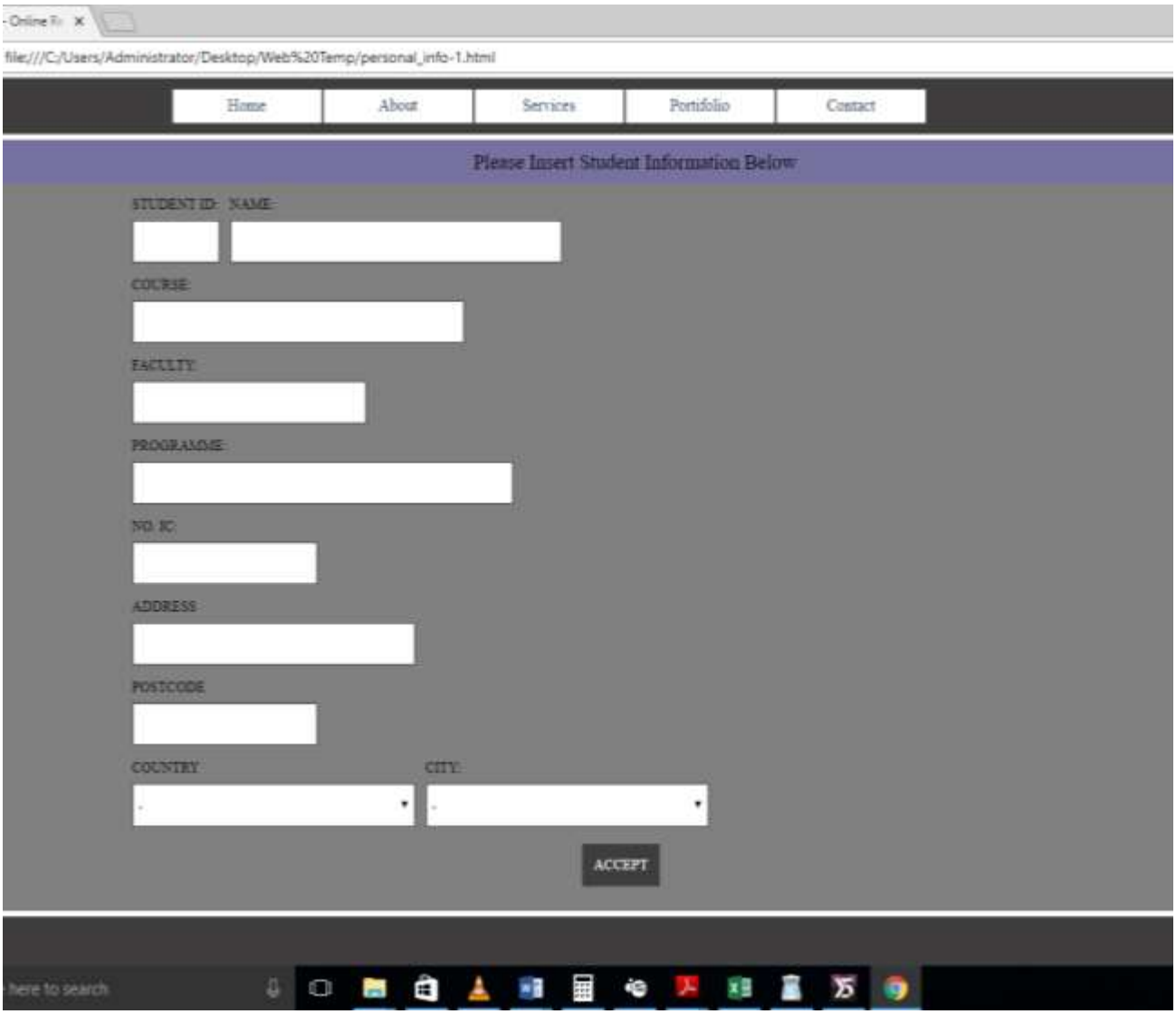


Figure 5: Personal Information form

Screen shoot for Accept Form

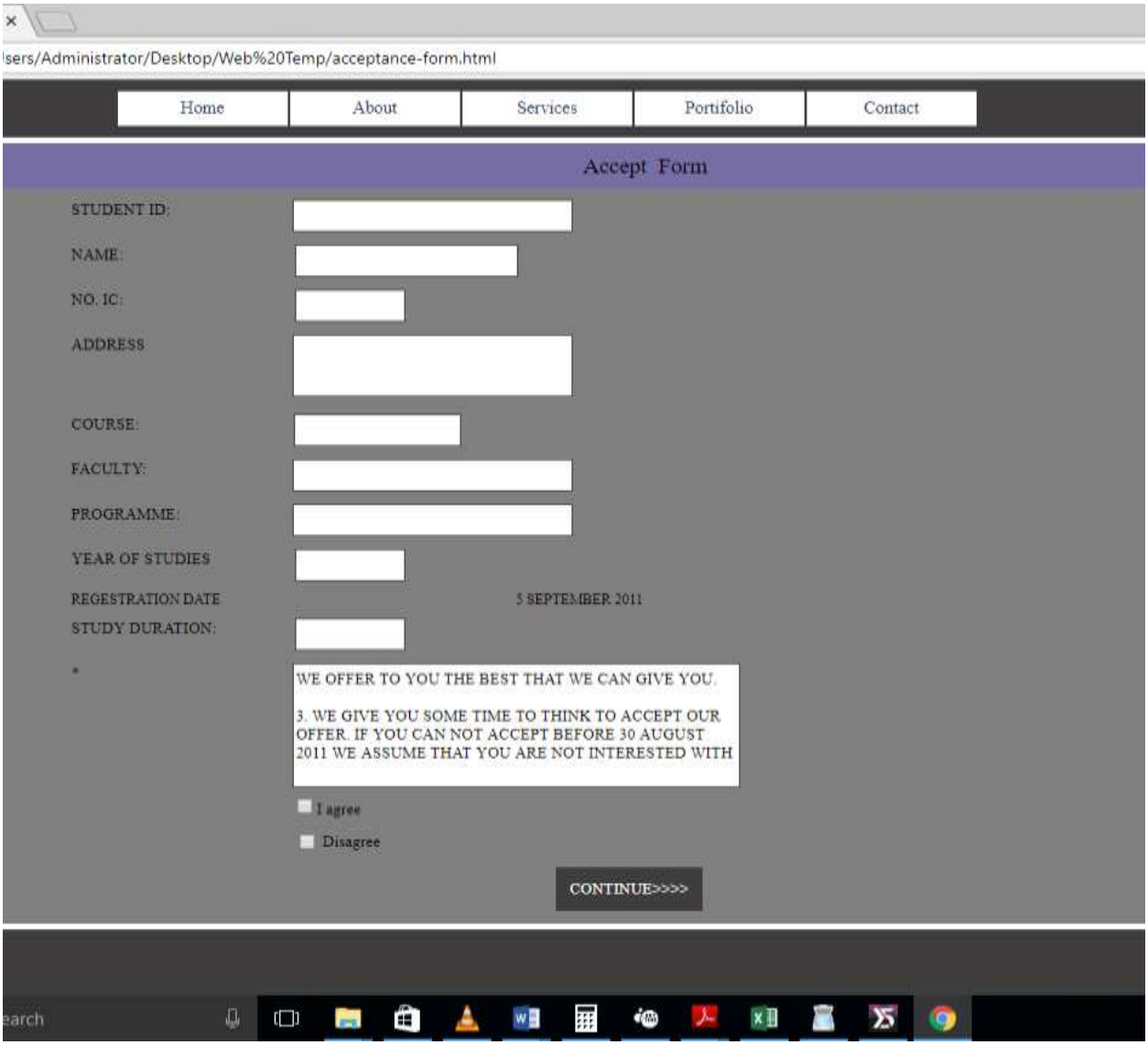


Figure 6: Accept Form

5.5. Results

The ORNSIS is expected to be able to help improve the current registration process conducted by library staff and students and able to help new students to register easily than manual system.

Among the expected outcomes from this project are:

a. Library administrator able to access to database to add new students' information.

b. Library administrator can able to log in into the system.

c. New students' can access to the system.

d. New students' can update their information.

e. Library administrator can get the actual number of students register online

f. Barcode scanner able to read the barcode at the offer letter.

5.6. Conclusion

The main focus of the Online Registration New Student Intake System (ORNSIS) for library registration is to produce an easy platform for both new student and library administrator in registration process. The aim is to retrieve actual number of student that registers as university students. This system also enables library administrator to monitor the progress of each new student who did the register online, did not register and reject the offer. Library administrator job needs to assign student who get the offer to register into database. The barcode for each student who get offer also stored in database. Overall, this system will reduce a lot of human energy and reduce the cost of registration process.

5.7. Requirement Analysis

The functional requirements of the proposed automated system will be calculations, technical details or specific functionality that shows how a user case is to be fulfilled. it will provide a layout of user cases for the system that will be supported by the non functional requirements of the design.

The user requirements of the proposed automated registration system must provide only authorized users with access to the system through user authentication and authorization to prevent manipulation of personal records. By putting the names user name password title of the book, ID number, faculty home or village name for need of information records.

5.8. Functional Requirements of the Automated Library User Registration System

These are key requirements which help the system to function without them; the system cannot meet its designed objectives .Functional requirements specify what the designed system must be able to do or accomplish in order to meet the library goals .Below is the summary of the functional requirements of the system;

- The system should be able to store, add edit, delete provide the details of the library user.
- It should also indicate missing users' information demand in the library.
- It should be able to report arrival and departure of library users and as well generate monthly reports by categorizing library users into educational levels, institutions age reasons to visit the library and others

5.9. Non-Functional Requirements

These capture the required properties of the system such as constraints, reliability and others. The following are considered non-functional requirements;

- The system should be user friendly
- It should be extended by other experts designing library system
- It should be able to add more features in the future
- The system should be error free in report generation to those who need to make more research; the system should provide controlled access to certain functions of the system library.

5.10. Hardware Requirement

The preferred system would need a computer with atleast400mb random access memory and should a keyboard. In addition to the system should have a colored monitor screen with a minimum processing speed, a printer preferably a laser.

5.11. Software Requirement

The application runs under window xp which should be pre-installed on the computer antivirus soft toolkit to be used is macfe antivirus that should be regularly upgraded and bar technology.

CHAPTER SIX: SUMMAR, CONCLUSION AND RECOMMENDATIONS

5.1. Introduction

This chapter presents summary, conclusion and recommendations.

5.2. SUMMARY

The topic of the project was designing an automated library user registration system for Nkumba university library.

5.2.1. Type of system used to register library users in Nkumba university library

Objective two was to examine the current system used in registering users in Nkumba university library and the findings from the respondents indicated that the registration system is highly manual where by too much paper work is involved during the registration process. According to the procedures and regulations, only users with proper identifications for example university identification card or appointment letters are eligible for membership and no one can be registered without such requirements.

During the study, several challenges were pointed out which include, pressure, too much paperwork, time consuming, data redundancy as illustrated in **table 4.3.1**

5.2.2 Automated User Registration System Design

Objective 5 of the study was to design an automated library user registration system. During the design, MSQL, PHP, HTML were used as the design**.**

5.3 CONCLUSION

Basing on the results of the study, 100% of the existing system is manual system. The purpose of this study was to establish the problems and challenges which currently exist in the work flow of traditional manual user registration system is general and in that Nkumba university library is particular ; a solution is illustrated which may be carried out to overcome and reduce the problems by introducing the automated user registration system.

However, the study revealed that 44.4% of the manual system is paper work and time consuming as some of the challenges associated with manual in Nkumba university library.

Therefore, during the system study, respondents were interviewed on the need of automated user registration system and they were able to point out need to save time, cut costs hence it the automated system is implemented, this will help to solve the challenges with the manual system like too much paper work and time consuming

5.4 RECOMMENDATIONS

Considering the responses from different respondents, the researcher was able to come up with the following recommendations;

5.4.1 Intensive Training on System Management and Use

Library staff and users should undergo training on this new system since it is automated and its new in the library, there is need for knowledge especially in automation and a system in order to effectively use services thus calls for training.

5.4.2 Data backup

The library management should ensure that there is proper data backup so that in case the system breaks down, the library can be in position to retrieve all the information**.**

5.4.3 Information security measures

Nkumba university library should put in place very serious measures as that to avoid data losses due to virus attack and system hackers. Such measures may include user authentication, installing anti-virus and data encryption.

5.4.4 Extension of other academic libraries

The system was designed and piloted at Nkumba university library .the system should therefore be extended to all school libraries in Uganda so as to harvest maximally the befits associated with it.

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